

Advantages:

• Can prove non-context-free languages
• Simple theoretical tool
• Pumping lemma ~~provides~~ ~~constructive~~ ~~useful~~
contradiction. Based on strong mathematical logic.

Disadvantages:

- Cannot prove a language is context-free
- Sometimes difficult to apply
- ~~Useful in some cases~~ Requires careful case analysis
 - Not always easy to find contradiction.

QED